

Radialization has BMO norm $\Theta(\sqrt{n})$, and symmetric rearrangement is not dimension-free

Solution packet for arXiv:2201.05130

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Status: substantial partial result, likely valid. The decreasing-rearrangement constant remains open. The packet completely classifies the dimension growth of the radialization step, proves that the symmetric-rearrangement constant grows at least as $\sqrt{n}$, and improves its known upper bound from exponential growth to $O(n)$.

1 Source question and scope

Burchard, Dafni, and Gibara ask for the sharp dimension dependence of the decreasing-rearrangement constants C_n and state that the sharp constants and dimension dependence for symmetric decreasing rearrangement are also unknown; see the introduction of arXiv:2201.05130, PDF pages 1–2 [1]. The full passage is reproduced below.

this is precluded by Sard's lemma; for the same reason, Steiner symmetrization is continuous in any dimension [7]. Similar questions have been studied regarding the boundedness and continuity of maximal functions on Sobolev spaces and BV [1, 10, 18, 19].

Turning to the space BMO of functions of bounded mean oscillation, it is well known that the decreasing rearrangement (that is, the map $f \mapsto f^*$) is bounded on BMO: there are constants C_n , depending only on dimension, such that

$$\|f^*\|_{\text{BMO}} \leq C_n \|f\|_{\text{BMO}}$$

whenever the decreasing rearrangement is defined. The sharp dependence of the constants C_n or, indeed, if they do depend on dimension at all is still an open question for dimensions $n > 1$. See [8] for a discussion of this inequality and for a proof that C_n exhibits at most square-root-dependence

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on n . It was proven in [8] that the symmetric decreasing rearrangement (that is, the map $f \mapsto Sf$) is also bounded on BMO. The sharp constants and their dependence on dimension remains unknown for $n \geq 1$.

For a cube Q and a locally integrable u , write

$$\mathcal{O}(u, Q) = \frac{1}{|Q|} \int_Q |u - u_Q|, \quad \|u\|_{\text{BMO}(\mathbb{R}^n)} = \sup_Q \mathcal{O}(u, Q),$$

where cubes have sides parallel to the coordinate axes. Let f^* be the nonincreasing rearrangement of $|f|$, let ω_n be the volume of the Euclidean unit ball, and set

$$Sf(x) = f^*(\omega_n|x|^n).$$

Denote by $\mathfrak{C}_n$ and $\mathfrak{D}_n$ the optimal constants in

$$\|f^*\|_{\text{BMO}(\mathbb{R}_+)} \leq \mathfrak{C}_n \|f\|_{\text{BMO}(\mathbb{R}^n)}, \quad \|Sf\|_{\text{BMO}(\mathbb{R}^n)} \leq \mathfrak{D}_n \|f\|_{\text{BMO}(\mathbb{R}^n)}.$$

The first source question—whether $\mathfrak{C}_n$ is dimension-free—is not settled here. The result answers the dimension-free part of the second question negatively and isolates the precise growth of the intervening radial operation.

2 Proof intuition

There are two independent mechanisms. First, a decreasing one-dimensional BMO function can change by at most a constant times its BMO norm whenever its argument is multiplied by two. Thus the oscillation of $g(\omega_n|x|^n)$ on a cube is controlled by the oscillation of $n \log|x|$. For a uniform point X in any cube, a Poincare estimate and a dimension-sharp small-ball bound give $\text{Var}(\log|X|) = O(1/n)$. Consequently radialization costs at most $O(\sqrt{n})$, and the logarithmic profile shows that this is sharp.

Second, to turn the logarithmic profile into the rearrangement of a function whose original BMO norm stays bounded, use the cusp

$$f_n(x_1, x') = (-\log|x_1| - \|x'\|_\infty)_+.$$

Its level sets have volume exactly $A_n e^{-t}$, so its symmetric rearrangement is a multiple of $-n \log|x|$. Yet on small cubes the two terms in the cusp each have bounded oscillation, while on large cubes the positive set is sufficiently thin to dilute the mean. On a cube of the form $a[1, 2]^n$, the radial output is governed by $\frac{n}{2} \log \sum_i X_i^2$; the central limit theorem produces a $\sqrt{n}$ lower bound with an explicit constant.

3 Main result

For a nonnegative nonincreasing $g \in \text{BMO}(\mathbb{R}_+)$ define

$$(T_n g)(x) = g(\omega_n|x|^n),$$

and let $\mathfrak{R}_n$ be the operator norm of T_n from this monotone cone to $\text{BMO}(\mathbb{R}^n)$. Finally set

$$\kappa := \frac{1}{7} \sqrt{\frac{17}{5\pi}} = 0.148616\dots$$

Theorem 1 (Radialization and symmetric-rearrangement bounds). *For $n \geq 3$,*

$$\mathfrak{R}_n \leq 32 \left(1 + \frac{n\omega_n^{1/n}}{\pi \log 2} \sqrt{\frac{n}{n-2}} \right) = O(\sqrt{n}), \tag{1}$$

whereas

$$\liminf_{n \rightarrow \infty} \frac{\mathfrak{R}_n}{\sqrt{n}} \geq \frac{e\kappa}{4} > 0. \tag{2}$$

Hence $\mathfrak{R}_n = \Theta(\sqrt{n})$.

Moreover, the optimal symmetric-rearrangement constants satisfy

$$\liminf_{n \rightarrow \infty} \frac{\mathfrak{D}_n}{\sqrt{n}} \geq \frac{\kappa}{4} > 0, \quad (3)$$

and

$$\mathfrak{D}_n \leq 2(1 + 2\sqrt{n-1})\mathfrak{R}_n = O(n). \quad (4)$$

In particular, symmetric decreasing rearrangement does not have a dimension-free cube-BMO bound. If $\sup_n \mathfrak{C}_n < \infty$, then $\mathfrak{D}_n = \Theta(\sqrt{n})$.

The factor $2(1 + 2\sqrt{n-1})$ in (4) is Theorem 1.1 of arXiv:2011.09111 (PDF page 2) [2]. Its published corollary used a comparison between cubes and annular sectors and gave the much larger upper bound $2(1 + 2\sqrt{n-1})n^{n/2}\omega_n$ [2, PDF pp. 3–4].

4 The radialization upper bound

We first record a logarithmic modulus that uses only one-dimensional BMO.

Lemma 2 (Dyadic modulus). *Let g be nonincreasing on $\mathbb{R}_+$ and let $K = \|g\|_{\text{BMO}(\mathbb{R}_+)}$. At Lebesgue points $s, t > 0$,*

$$|g(s) - g(t)| \leq 16K \left(1 + \frac{|\log(s/t)|}{\log 2} \right).$$

Proof. It suffices first to compare a and $2a$. Let U, V be independent and uniform on $I = (a/2, 5a/2)$. On a set of probability $1/4$, U lies in $(a/2, a)$ and $g(U) \geq g(a)$; on another set of probability $1/4$, V lies in $(2a, 5a/2)$ and $g(V) \leq g(2a)$. Including the reversed event,

$$\mathbb{E}|g(U) - g(V)| \geq \frac{1}{8}(g(a) - g(2a)).$$

On the other hand, $\mathbb{E}|g(U) - g(V)| \leq 2\mathcal{O}(g, I) \leq 2K$. Therefore $g(a) - g(2a) \leq 16K$. A dyadic telescoping from s to t proves the claim. $\square$

Lemma 3 (Logarithmic radius in a cube). *Let X be uniform in an arbitrary cube of side length ℓ in $\mathbb{R}^n$, $n \geq 3$. Then*

$$\text{Var}(\log |X|) \leq \frac{1}{\pi^2} \frac{n}{n-2} \omega_n^{2/n}. \quad (5)$$

Proof. The Neumann Poincaré inequality on a cube gives, first for a smooth regularization of the logarithm and then by approximation,

$$\text{Var}(\log |X|) \leq \frac{\ell^2}{\pi^2} \mathbb{E}|X|^{-2}.$$

For every $r > 0$, the elementary volume bound

$$\mathbb{P}\{|X| < r\} \leq \min \left(1, \frac{\omega_n r^n}{\ell^n} \right)$$

holds regardless of the location of the cube. Put $r_0 = \ell \omega_n^{-1/n}$. Layer-cake integration yields

$$\begin{aligned}\mathbb{E}|X|^{-2} &= 2 \int_0^\infty r^{-3} \mathbb{P}\{|X| < r\} dr \\ &\leq \frac{2\omega_n}{\ell^n} \int_0^{r_0} r^{n-3} dr + 2 \int_{r_0}^\infty r^{-3} dr = \frac{n}{n-2} \frac{\omega_n^{2/n}}{\ell^2}.\end{aligned}$$

This proves (5). $\square$

Proof of (1). Fix a cube Q and let X be uniform on it. Set $S = \omega_n |X|^n$ and $s_0 = \exp(\mathbb{E} \log S)$. Lemma 2, followed by the elementary inequality $\mathcal{O}(Y) \leq 2\mathbb{E}|Y - c|$, gives

$$\begin{aligned}\mathcal{O}(T_n g, Q) &\leq 2\mathbb{E}|g(S) - g(s_0)| \\ &\leq 32K \left(1 + \frac{1}{\log 2} \mathbb{E}|\log S - \mathbb{E} \log S| \right) \\ &\leq 32K \left(1 + \frac{n}{\log 2} \sqrt{\text{Var}(\log |X|)} \right).\end{aligned}$$

Insert Lemma 3 and take the supremum over Q . Finally, Stirling's formula gives $\omega_n^{1/n} \sim \sqrt{2\pi e/n}$, so the displayed bound is $O(\sqrt{n})$. $\square$

5 Sharpness of radialization

We use the logarithmic profile

$$g_0(s) = (-\log s)_+.$$

For completeness, $\|g_0\|_{\text{BMO}(\mathbb{R}_+)} \leq 4/e$. Indeed, truncation does not increase mean oscillation. If an interval I of length ℓ lies at distance at least ℓ from zero, then $-\log s$ varies by at most $\log 2$ on I . Otherwise $I \subset (0, 2\ell)$, the containing interval has at most twice its length, and direct integration gives $\mathcal{O}(-\log s, (0, 2\ell)) = 2/e$.

Now

$$T_n g_0(x) = (-\log(\omega_n |x|^n))_+ = n \log(R_n/|x|)_+, \quad R_n = \omega_n^{-1/n}.$$

Take $a_n = R_n/(3\sqrt{n})$ and $Q_n = a_n[1, 2]^n$. This cube lies inside $B(0, R_n)$. If $X_1, \dots, X_n$ are independent and uniform on $[1, 2]$, then addition of constants and scaling show

$$\mathcal{O}(T_n g_0, Q_n) = \mathbb{E} \left| \frac{n}{2} \log \left(\sum_{i=1}^n X_i^2 \right) - \mathbb{E} \frac{n}{2} \log \left(\sum_{i=1}^n X_i^2 \right) \right|. \quad (6)$$

Let $Z_i = X_i^2$. A direct calculation gives

$$\mu := \mathbb{E} Z_i = \frac{7}{3}, \quad \sigma^2 := \text{Var}(Z_i) = \frac{34}{45}.$$

The delta method applied to $\sum_i Z_i$, together with boundedness of the Z_i (which supplies uniform integrability), implies that

$$\frac{1}{\sqrt{n}} \left[\frac{n}{2} \log \left(\sum_i Z_i \right) - \mathbb{E} \frac{n}{2} \log \left(\sum_i Z_i \right) \right] \rightarrow N \left(0, \frac{\sigma^2}{4\mu^2} \right)$$

in distribution and in first absolute moment. Hence the expression in (6), divided by $\sqrt{n}$, tends to

$$\frac{\sigma}{2\mu} \mathbb{E}|N(0, 1)| = \frac{\sigma}{\mu\sqrt{2\pi}} = \kappa.$$

Since $\|g_0\|_{\text{BMO}} \leq 4/e$, this proves (2).

6 A uniformly BMO logarithmic cusp

Define, for $n \geq 2$ and $x' = (x_2, \dots, x_n)$,

$$f_n(x_1, x') = (-\log |x_1| - \|x'\|_\infty)_+. \quad (7)$$

Lemma 4 (Uniform source norm). *f_n is rearrangeable and $\|f_n\|_{\text{BMO}(\mathbb{R}^n)} \leq 4$ for every $n \geq 2$.*

Proof. We first use the dimension-free bound

$$\|\log |\cdot|\|_{\text{BMO}(\mathbb{R})} \leq \frac{8}{e}. \quad (8)$$

If an interval I of length ℓ has distance at least ℓ from zero, the oscillation is at most the range $\log 2$. Otherwise $I \subset (-2\ell, 2\ell)$. Mean oscillation on a subset is at most the volume ratio times the oscillation on the containing interval, while

$$\mathcal{O}(-\log |t|, (-2\ell, 2\ell)) = \int_0^1 |-\log s - 1| ds = \frac{2}{e}.$$

The volume ratio is 4, proving (8).

Let $Q = I_1 \times \dots \times I_n$ have side length ℓ . If $\ell \leq 1$, truncation and the triangle inequality for mean oscillation give

$$\mathcal{O}(f_n, Q) \leq \mathcal{O}(-\log |x_1|, Q) + \mathcal{O}(\|x'\|_\infty, Q) \leq \frac{8}{e} + \ell < 4,$$

because $\|x'\|_\infty$ varies by at most ℓ on Q .

If $\ell > 1$, nonnegativity gives $\mathcal{O}(f_n, Q) \leq 2|Q|^{-1} \int_Q f_n$. For fixed x' and $v = \|x'\|_\infty$,

$$\int_{I_1} (-\log |x_1| - v)_+ dx_1 \leq \int_{|x_1| < e^{-v}} (-\log |x_1| - v) dx_1 = 2e^{-v}.$$

Consequently $|Q|^{-1} \int_Q f_n \leq 2/\ell$, and $\mathcal{O}(f_n, Q) < 4$. The level-set computation below also shows directly that f_n is rearrangeable. $\square$

For every $t \geq 0$, Fubini's theorem and the volume of an ℓ_∞ -ball give

$$\begin{aligned} |\{f_n > t\}| &= 2e^{-t} \int_{\mathbb{R}^{n-1}} e^{-\|x'\|_\infty} dx' \\ &= 2e^{-t} \int_0^\infty e^{-r} d(2r)^{n-1} = A_n e^{-t}, \quad A_n = 2^n (n-1)!. \end{aligned}$$

It follows exactly that

$$f_n^*(s) = \log(A_n/s)_+, \quad Sf_n(x) = n \log(\rho_n/|x|)_+, \quad \rho_n = (A_n/\omega_n)^{1/n}.$$

Taking the cube $\rho_n(3\sqrt{n})^{-1}[1, 2]^n$ repeats the calculation (6); hence

$$\liminf_{n \rightarrow \infty} \frac{\|Sf_n\|_{\text{BMO}}}{\sqrt{n}} \geq \kappa.$$

Lemma 4 proves (3).

Finally, $Sf = T_n(f^*)$, so by definition $\mathfrak{D}_n \leq \mathfrak{R}_n \mathfrak{C}_n$. Theorem 1.1 of [2] gives $\mathfrak{C}_n \leq 2(1 + 2\sqrt{n-1})$, proving (4) and the conditional $\Theta(\sqrt{n})$ statement.

7 Verification and limitations

The analytic proof was checked in four independent pieces: the dyadic BMO modulus, the layer-cake calculation in Lemma 3, the exact cusp distribution, and the delta-method constant. The seeded script `code/verify_asymptotics.py` sampled 200,000 vectors for each of $n = 10, 25, 50, 100, 200$. The normalized values were respectively

$$0.15054, 0.14948, 0.14889, 0.14830, 0.14871,$$

against the proved limit $\kappa = 0.148616\dots$. This computation is a sanity check only; the proof is the central limit theorem plus uniform integrability.

A separate finite-grid search for a growing lower bound on $\mathfrak{C}_n$ was rejected: ratios above one disappeared after arbitrary shifted cubes were included. It supplies no evidence or premise used here. The present result leaves the sharp growth of $\mathfrak{C}_n$ open and leaves a gap $\sqrt{n} \lesssim \mathfrak{D}_n \lesssim n$. Closing the upper gap by the factorization $Sf = T_n(f^*)$ is equivalent to obtaining a dimension-free bound for $\mathfrak{C}_n$.

Bounded novelty search. On 17 August 2026, searches of arXiv/web used ids 2201.05130 and 2011.09111 and the phrases “symmetric decreasing rearrangement BMO dimension lower bound”, “radially decreasing BMO cube norm”, “logarithmic cusp”, and “symmetrization BMO dimension-free”. They returned the two cited source papers and unrelated rearrangement literature, but no later match for (1)–(4). This is bounded evidence, not an exhaustive claim of novelty.

Human-review recommendation. Priority checks are: (i) the use of cube Poincare after regularizing $\log|x|$ at the origin; (ii) the truncation and large-cube estimates in Lemma 4; and (iii) uniform integrability in the delta-method limit. These are the only points where an omitted technical detail could affect the theorem. No computational dependency remains.

References

- [1] A. Burchard, G. Dafni, and R. Gibara, *Vanishing Mean Oscillation and Continuity of Rearrangements*, arXiv:2201.05130 (2022), especially introduction, PDF pp. 1–2.
- [2] A. Burchard, G. Dafni, and R. Gibara, *Mean Oscillation Bounds on Rearrangements*, Trans. Amer. Math. Soc. 375 (2022), 4429–4444; arXiv:2011.09111, Theorems 1.1–1.2.